

Lesson 13 — Sampling & Sampling Distribution

20 Practice Exercises (with Python)

Based on: Population vs Sample · Sampling Methods · Sample Statistics · Sampling Distribution · Central Limit Theorem (CLT) · Standard Error · Bias & Variance

Part 1 — Population vs Sample (Conceptual)

Exercise 1. Explain in your own words the difference between a **population** and a **sample**. Give one real-life example (different from the lecture's "student height" example) where measuring the whole population would be too expensive or impossible.

Exercise 2. Match each symbol to what it represents, and say whether it belongs to the population or the sample: μ , σ , $\bar{x}$, s , N , n

Exercise 3. A company wants to know the average salary of all 3,000,000 workers in a country. They survey 500 randomly chosen workers instead. Identify:

- What is N ? What is n ?
- What is unknown here (population parameter)?
- What is calculated here (sample statistic)?

Exercise 4. Why do we say $\bar{x}$ and s are used to **estimate** μ and σ , rather than being equal to them? Explain with one sentence about sampling error.

Part 2 — Sampling Methods (Simple Random, Stratified, Cluster, Systematic)

Exercise 5. For each scenario below, choose the correct sampling method (Simple Random, Stratified, Cluster, or Systematic) and justify your choice:

- a) A school has 5 grades. You want 20 students from each grade.
- b) You pick every 10th customer entering a supermarket.
- c) You randomly select 8 villages out of 200, then survey every household in those 8 villages.
- d) You have a list of 10,000 emails and pick 200 completely at random.

Exercise 6. Using Python and numpy, generate a population of 1,000 random integers between 150 and 200 (simulating heights in cm). Then draw a **Simple Random Sample** of size 50 without replacement.

```
import numpy as np
population = np.random.randint(150, 200, size=1000)
# your code: draw a simple random sample of size 50
```

Exercise 7. Create a pandas DataFrame with 200 students, each assigned to one of 4 “kurs” (year) groups (1–4), with a random bo'y (height) column. Perform **Stratified Sampling**: select 10 students from each kurs group.

```
import pandas as pd
import numpy as np
# your code here
```

Exercise 8. Simulate a population of 100 elements split into 20 clusters of 5 elements each. Randomly select 4 clusters (**Cluster Sampling**) and print all elements belonging to those clusters.

Exercise 9. Given a population list of 300 elements, implement **Systematic Sampling** to select a sample of size 15 (i.e., calculate $k = N // n$ and take every k-th element).

Exercise 10. Compare Simple Random Sampling and Systematic Sampling: name one advantage and one disadvantage of each.

Part 3 — Sample Statistics ($\bar{x}$, s^2 , s)

Exercise 11. Given the sample [12, 15, 14, 10, 13, 16, 11], calculate by hand (show your work): - Sample mean $\bar{x}$ - Sample variance s^2 (using $n-1$, Bessel's correction) - Sample standard deviation s

Exercise 12. Verify your Exercise 11 answers using Python:

```
import numpy as np
sample = [12, 15, 14, 10, 13, 16, 11]
# calculate mean, variance (ddof=1), std (ddof=1)
```

Exercise 13. Explain why `np.var(sample, ddof=1)` is used instead of `np.var(sample, ddof=0)` when working with a sample rather than a full population. What does “Bessel's correction” fix?

Exercise 14. Write a Python function `sample_stats(sample)` that returns a dictionary with mean, variance, and std for any given list of numbers, without using numpy (compute manually using loops/formulas).

Part 4 — Sampling Distribution & Central Limit Theorem (CLT)

Exercise 15. In your own words, define **Sampling Distribution of the Mean**. What are its two key parameters, and how are they related to the population's μ and σ ?

Exercise 16. Simulate the Sampling Distribution described in the lecture: - Create a population with $\mu = 170$, $\sigma = 10$ (e.g., `np.random.normal(170, 10, 100000)`) - Draw 1,000 samples, each of size $n = 50$ - Compute the mean of each sample (you should

get 1,000 $\bar{x}$ values) - Plot a histogram of these 1,000 means - Compare the mean and std of your 1,000 $\bar{x}$ values to the theoretical μ and $\sigma/\sqrt{n}$

```
import numpy as np
import matplotlib.pyplot as plt
# your simulation code here
```

Exercise 17. Repeat Exercise 16, but start with a **non-normal population** (e.g., an exponential or uniform distribution) instead of a normal one. Show that the distribution of sample means still looks approximately normal when $n \geq 30$. This demonstrates the Central Limit Theorem (CLT).

Exercise 18. The CLT states $\bar{x} \sim N(\mu, \sigma^2/n)$. If a population has $\sigma = 20$ and you take samples of size $n = 25$, what is the theoretical standard deviation of the sampling distribution (i.e., the Standard Error)? Show the formula and the calculation.

Part 5 — Standard Error, Bias & Variance

Exercise 19. Given $\sigma = 10$, calculate the Standard Error $SE = \sigma/\sqrt{n}$ for $n = 1, 4, 25, 100, 400$. Then answer: if you want to cut the Standard Error in half, by what factor must you increase the sample size n ? Explain why this is called the “quadratic cost” of precision.

Exercise 20. Two estimators were used to estimate a population mean $\mu = 50$ across many repeated samples: - Estimator A: average of $\bar{x}$ values = 50.1, with high spread (variance) between samples - Estimator B: average of $\bar{x}$ values = 55.0, with low spread (variance) between samples

For each estimator, determine: - Its approximate **Bias** ($Bias = E[\bar{x}] - \mu$) - Whether it has **low or high variance** - Which quadrant of the “bias vs variance” target diagram it belongs to (Low Bias/Low Var, High Bias/Low Var, Low Bias/High Var, High Bias/High Var) - Which estimator would you trust more overall, and why?

Quick Reference (Formulas)

Concept	Formula
Sample mean	$\bar{x} = (1/n) \sum x_i$
Sample variance	$s^2 = \sum (x_i - \bar{x})^2 / (n-1)$
Sample std	$s = \sqrt{s^2}$
CLT	$\bar{x} \sim N(\mu, \sigma^2/n)$ for $n \geq 30$
Standard Error	$SE = \sigma/\sqrt{n}$
Bias	$Bias = E[\bar{x}] - \mu$
Variance of $\bar{x}$	$Var[\bar{x}] = \sigma^2/n$

Tip: Do exercises 1–14 first (concepts + basic stats), then 15–20 (simulation-heavy, requires understanding of CLT). Try to run every code exercise yourself in Python/Jupyter rather than just reading it.